

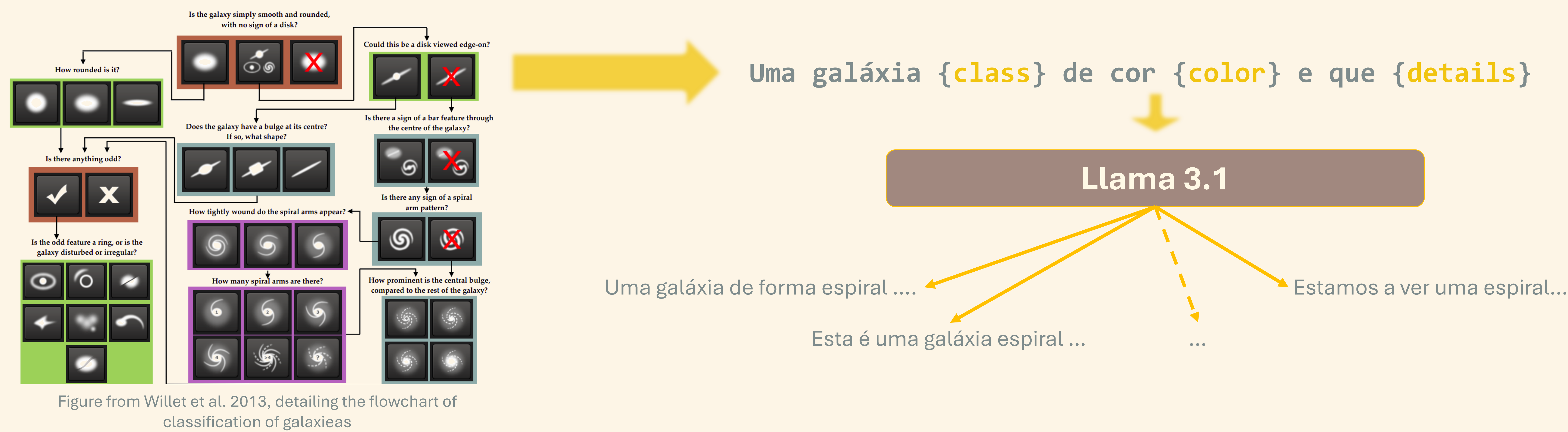
AstroLingua: Empowering Inclusive Astronomy through AI

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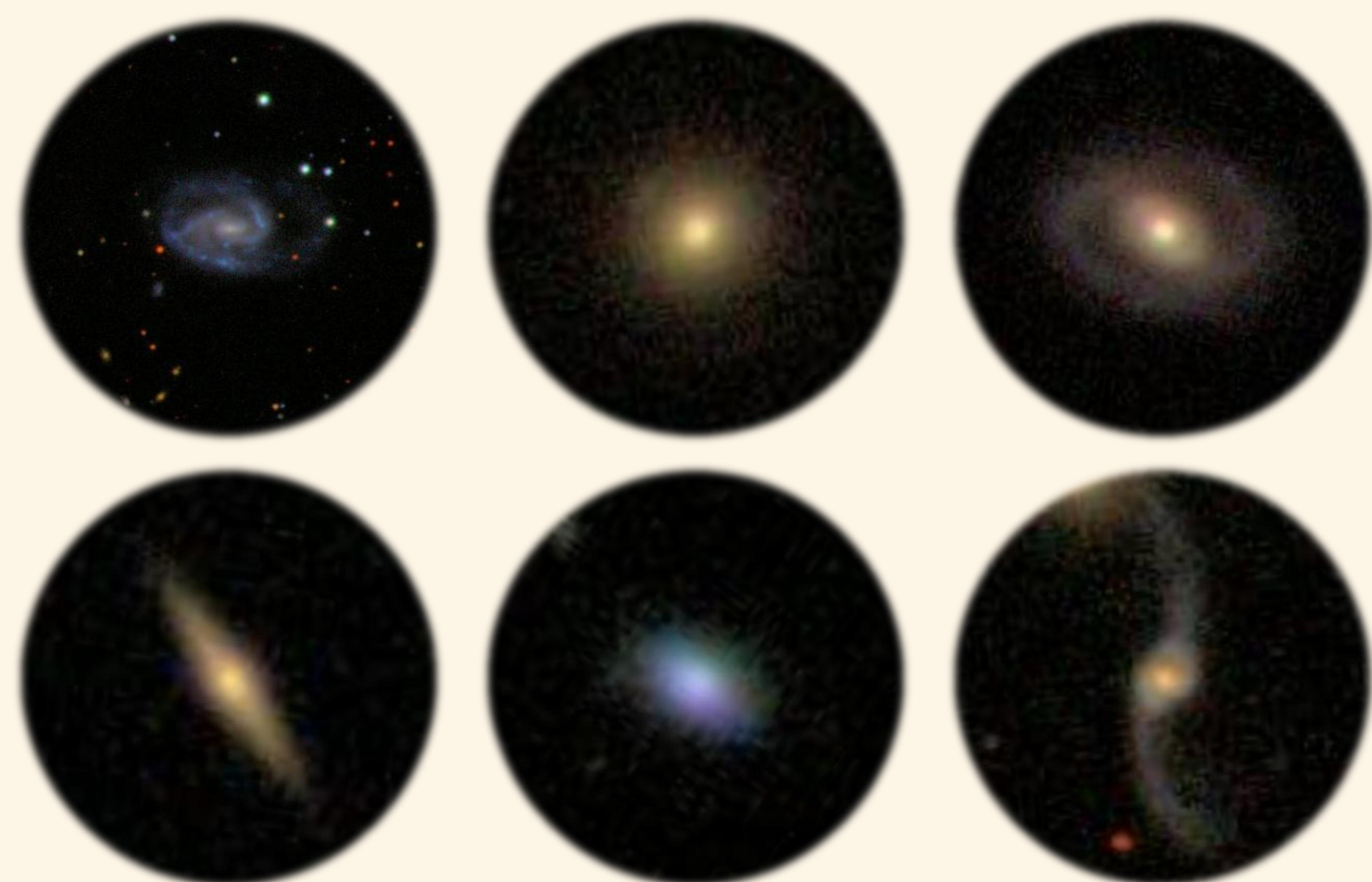


Images of extragalactic objects have captured the fascination of the public, giving rise to a new generation of citizen science projects. The Galaxy Zoo project, gathering nearly one million classifications worldwide, provides a diverse dataset of galaxies in various shapes and sizes and cosmic times. The **AstroLingua** project aims at developing a comprehensive pipeline that seamlessly integrates state-of-the-art image analysis techniques and natural language processing (NLP) to deliver detailed descriptions of galaxies based on their morphology and environmental characteristics.

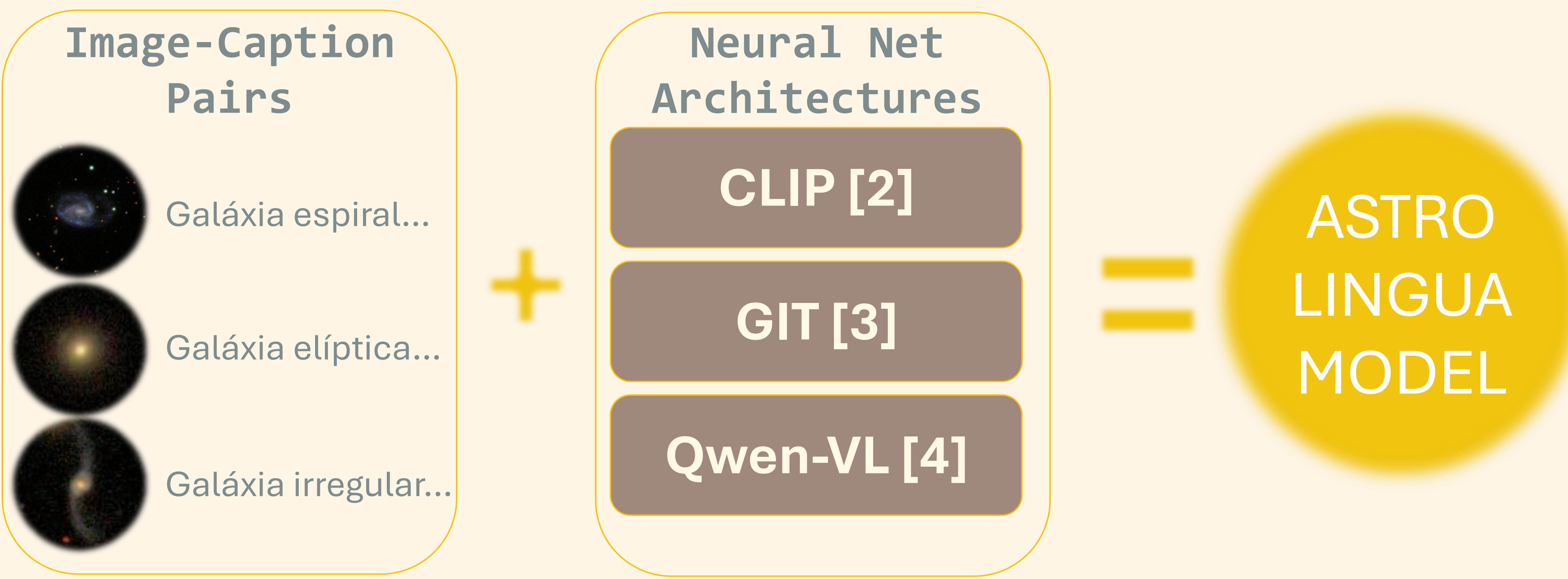
1. Use Galaxy Zoo [1] classifications to generate base template captions.
Then use a generic LLM to create variations for data augmentation



2. Collect Images from Galaxy Zoo SDSS

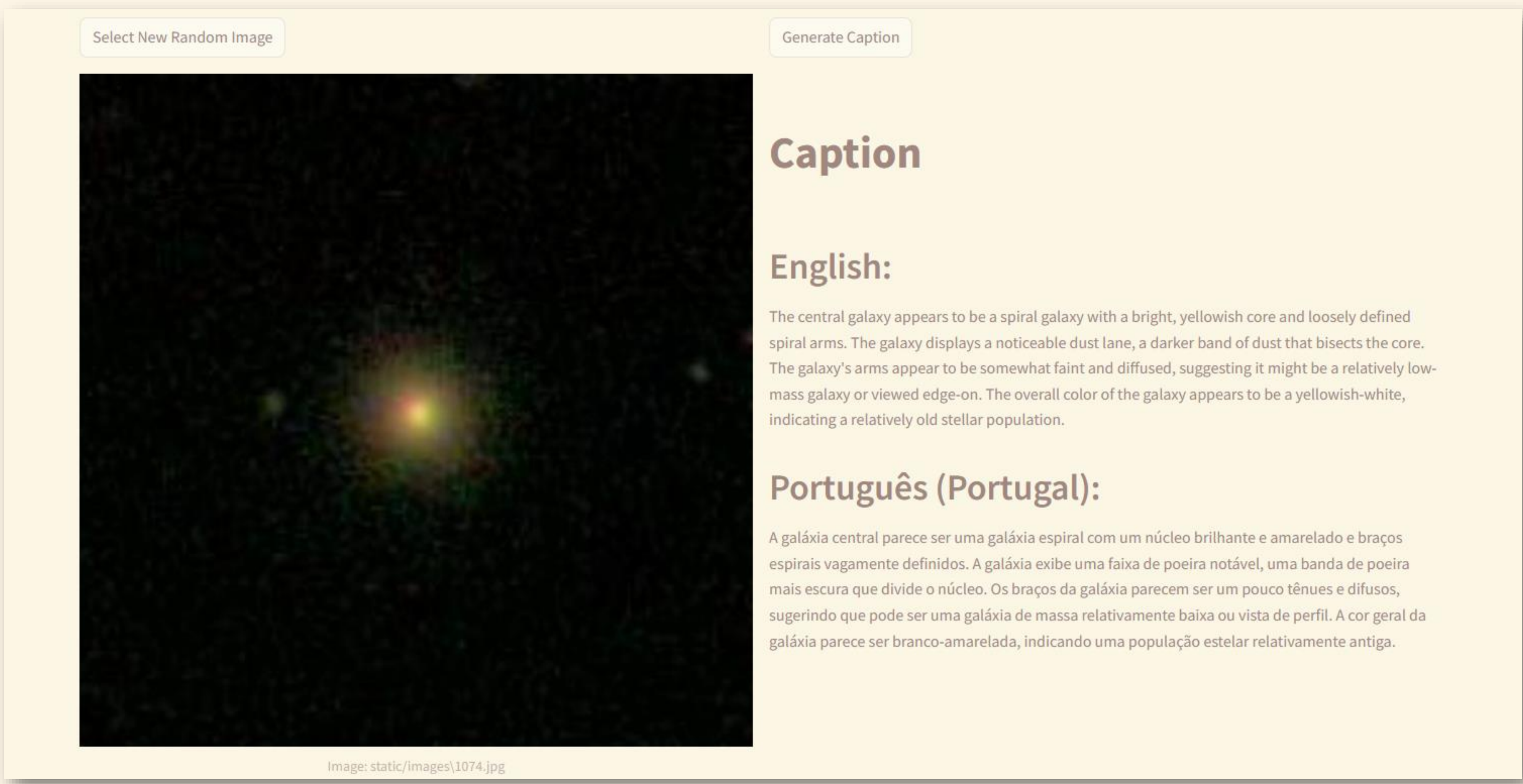
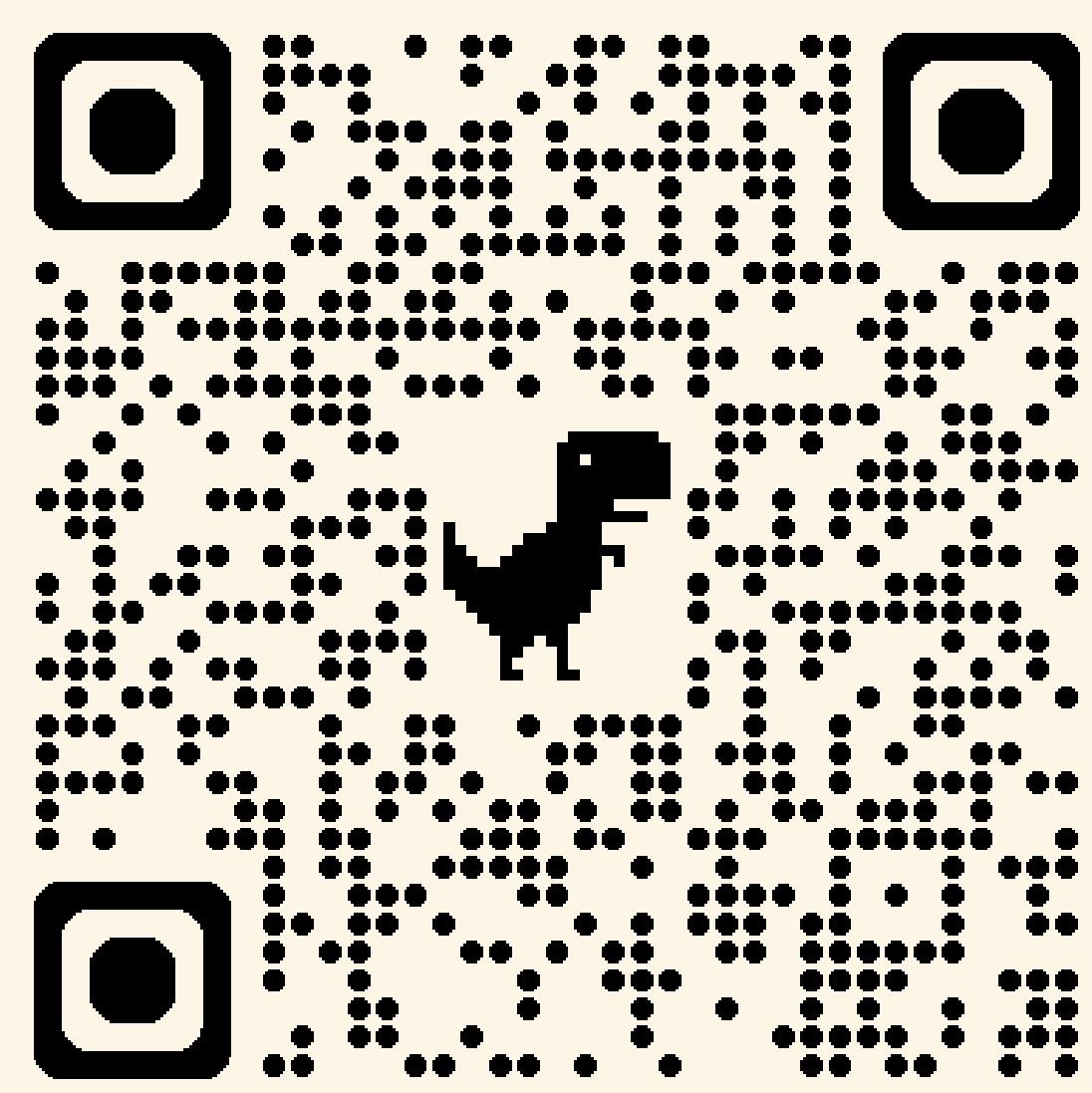


3. Fine-tune Image-to-text model



4. Make the model publicly available for download and through a web application for dissemination

Try it yourself here



REFERENCES: [1] Willett K. W. et al., 2013, MNRAS, 435, 2835; [2] Radford A. et al., 2021, arXiv, arXiv:2103.00020; [3] Wang J. et al., 2022, arXiv, arXiv:2205.14100; [4] Bai J. et al., 2023, arXiv, arXiv:2308.12966

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